

LECTURE 7.2

Fundamental PDEs

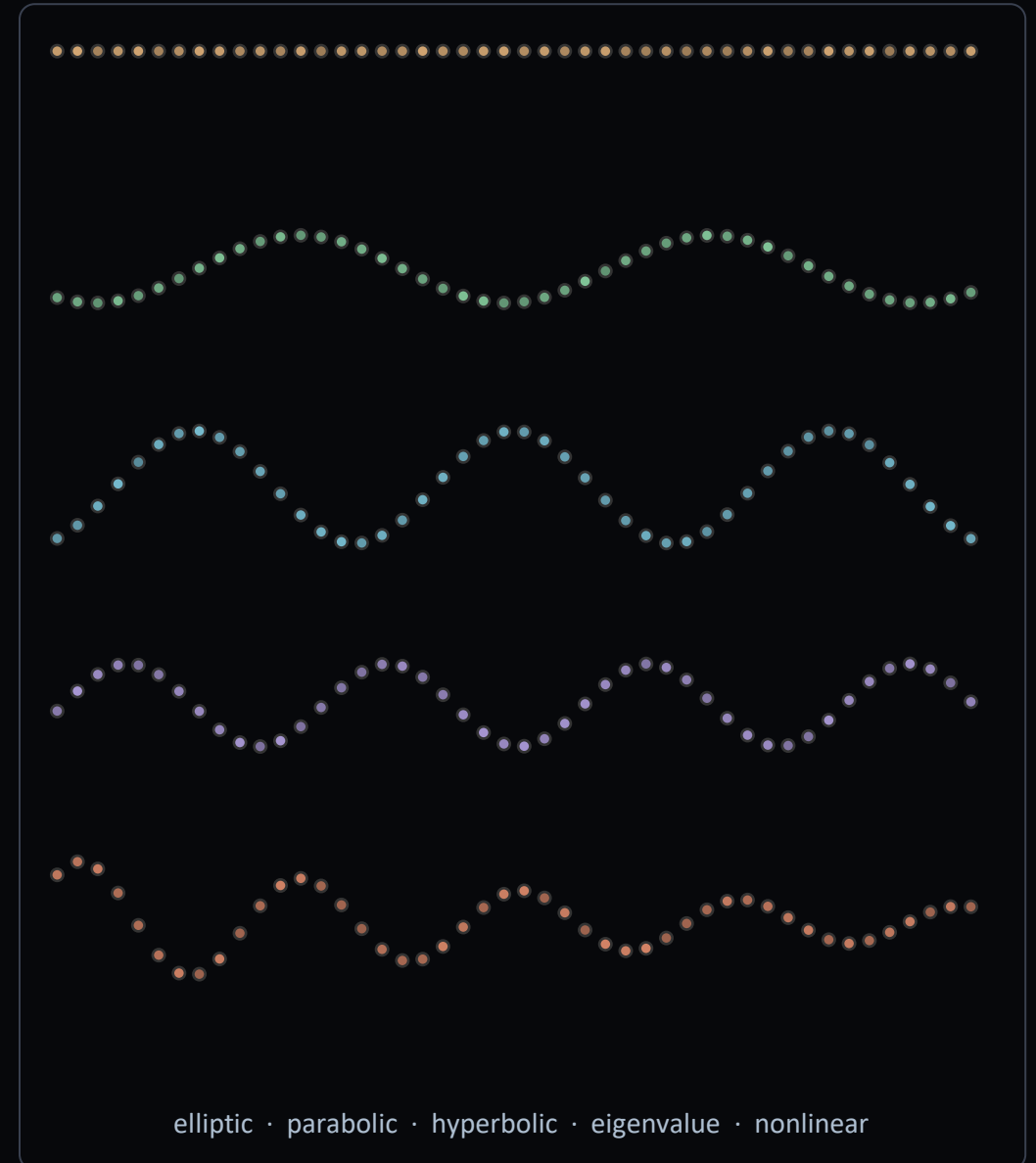
The mathematical backbone for fluid flow, heat dissipation, electrochemical devices, power grids and autonomous robots

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Aalborg University · ret@et.aau.dk · Deep Learning for Engineering · 2026

Liu (2025), PINN with Python, Ch. 2–6 · Raissi, Perdikaris & Karniadakis (2019)



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **classify** — a PDE as elliptic, parabolic or hyperbolic, and say what follows
- **count** — how many conditions a problem needs, and recognise one that is short
- **build** — a trial solution that enforces an initial condition exactly
- **explain** — why the wave equation needs two initial conditions, physically
- **choose** — an activation from the character of the solution, not from habit
- **recognise** — the trivial-solution trap, and say what excludes it

WHAT TODAY ASSUMES

from L7.1	residual, composite loss, hard enforcement
from L7.1	the T3 quadrature machinery
from L4.1	why the activation must be smooth

ONE PIPELINE, FIVE EQUATIONS

Nothing in L7.1's five stages changes today. Only the operator and the admissible conditions do.

AND ONE NEW KIND OF FAILURE

Helmholtz is the first problem where the loss must exclude a solution rather than attract one.

What Carries Over From L7.1 — and What Changes

UNCHANGED FOR EVERY PDE

- ✓ The network is still the solution ansatz, $u \approx \mathbb{N}\hat{w}$.
- ✓ Derivatives still come from autograd, exactly.
- ✓ The loss is still a sum of squared residuals.
- ✓ Training is still Adam followed by L-BFGS.
- ✓ Sampling still obeys $N_{\text{sample}} \geq P^*$.

CHANGES PER PDE

- The operator $\mathcal{F}$ — which derivatives appear, and how.
- Which admissible conditions exist: DBC, NBC, IC, IV.
- Whether a trial solution can hard-enforce them.
- Which activation suits the solution's character.
- Where the sampling has to be dense.

EVERYTHING IN THIS LECTURE IS ONE INSTANCE OF THIS

$$\mathcal{F}(u; \mathbf{x}, t, \mathbf{c}) = 0, \quad \mathbf{x} \in \Omega, \quad t \in [0, t_{\text{end}}]$$

Five PDEs follow, ordered so that each introduces exactly one new mathematical difficulty. Nothing is repeated for its own sake.

Three Types, One Discriminant

GENERAL SECOND-ORDER LINEAR PDE IN TWO VARIABLES

$$A u_{,xx} + B u_{,xy} + C u_{,yy} + D u_{,x} + E u_{,y} + F u = G$$

DISCRIMINANT

$$\Delta = B^2 - 4AC$$

POISSON

Elliptic

$$\Delta < 0$$

No time. Equilibrium: every point feels every boundary at once. Smooth solutions.

HEAT

Parabolic

$$\Delta = 0$$

First order in time. Diffusive: disturbances smooth out and spread instantly.

WAVE

Hyperbolic

$$\Delta > 0$$

Second order in time. Information travels at finite speed along characteristics.

The type dictates which conditions make the problem well posed — and therefore which loss terms a PINN must carry. Helmholtz and Burgers extend the picture: an eigenvalue problem and a nonlinear one.

The Five PDEs at a Glance

PDE	TYPE	OPERATOR $\mathcal{F}$	CONDITIONS	NEW DIFFICULTY
Poisson	<i>elliptic</i>	$\nabla^2 u + f = 0$	DBC, NBC	hard enforcement
Transient heat	<i>parabolic</i>	$u_t = c \nabla^2 u + f$	DBC, IC	space–time domain
Wave	<i>hyperbolic</i>	$u_{tt} = c^2 \nabla^2 u + f$	DBC, IC, IV	second initial condition
Helmholtz	<i>eigenvalue</i>	$\nabla^2 u + \lambda u = 0$	DBC + normalisation	λ is unknown too
Burgers	<i>nonlinear</i>	$u_t + uu_x = \nu u_{xx}$	IC, DBC	shock formation

Read the last column downwards: a source term, then time, then second-order time, then an unknown eigenvalue, then nonlinearity. That is the whole lecture.

Poisson's Equation: the Reference Problem

STRONG FORM

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + f(x, y) = 0$$

- **the simplest PDE that is still genuinely one** — second order, two dimensions, no time
- any steady-state balance with a distributed source
- **no time means no initial condition** — the boundary alone determines the solution
- only second spatial derivatives, so tanh and two autograd passes suffice

WHERE IT APPEARS IN THIS COURSE

Steady heat conduction

L8

temperature in a plate or
heat sink with internal load

Electrostatics

L9

potential from a charge
distribution in a device

Incompressible flow

L9

the pressure equation inside
a Navier–Stokes solver

DC power flow

L11

the linearised grid model
over a network Laplacian

Boundary Conditions and Well-Posedness

DIRICHLET ON ONE PART OF THE BOUNDARY, NEUMANN ON THE REST

$$u \Big|_{\partial\Omega_{DBC}} = \bar{u}, \quad \frac{\partial u}{\partial n} \Big|_{\partial\Omega_{NBC}} = \bar{t}$$

- well posed when a condition is prescribed at every boundary point
- **Dirichlet, Neumann, or a mix** — but no gaps and no double prescription
- **a pure Neumann problem is special** — determined only up to an additive constant
- physically: fix the state, or fix the flux

A MIXED PROBLEM

$u = \bar{u}$



$\partial u / \partial n = \bar{t}$

solid: Dirichlet · dashed: Neumann

the two must together cover all of $\partial\Omega$, without overlap

Hard Enforcement on a Rectangle

TRIAL SOLUTION ON $[A, B] \times [C, D]$

$$u_{trial}(x, y) = x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}(x, y)$$

- the polynomial multiplier vanishes on all four edges
- so the Dirichlet condition holds identically, before any training
- the boundary loss disappears and the composite loss collapses to one term
- for a non-zero boundary value, add it back
- **measured in L7.1** — about 165× more accurate, at the same cost

THE ENTIRE LOSS

$$\mathcal{L} = \mathcal{L}_{PDE}$$

THE CATCH

The multiplier must be constructed for the geometry. A rectangle is easy; an arbitrary domain is not — which is the subject of the next slide, and the reason Chapter 3 exists.

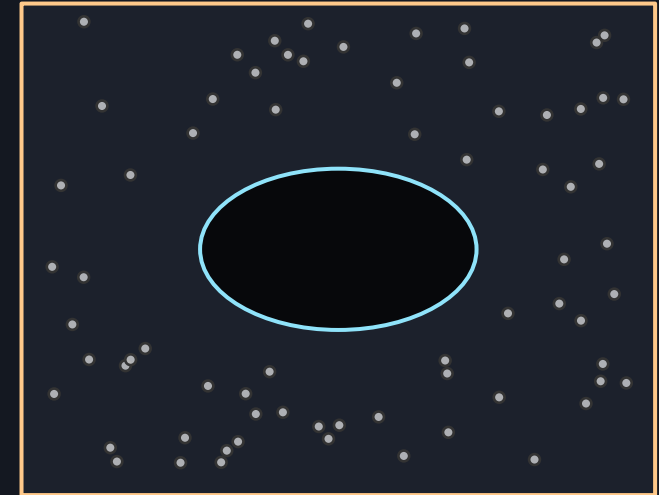
Irregular Domains: Learning the Boundary Function

TRIAL SOLUTION WITH A LEARNED MULTIPLIER

$$u_{trial}(x, y) = \mathcal{N}_{\hat{v}}^{DBC}(x, y) \mathcal{N}_{\hat{w}}(x, y)$$

- on a domain with no closed-form boundary polynomial, train a small second network
- then freeze it and use it exactly as the polynomial was
- **the domain is discretised into a T3 mesh** — L7.1's quadrature machinery
- which is what makes PINNs usable on geometries that actually occur

PLATE WITH AN ELLIPTICAL HOLE



$\partial\Omega$ outer: Dirichlet
hole: Neumann (traction-free)

The multiplier must vanish on the outer edge and stay positive around the hole — no single polynomial does this, so it is learned.

Transient Heat: Adding Time

GOVERNING EQUATION

$$\frac{\partial u}{\partial t} = c \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + f(x, y, t)$$

- **one new term, the time derivative** — and a third input to the network
- everything else is Poisson
- c is the thermal diffusivity, and an obvious candidate for an inverse problem
- first order in time, so exactly one initial condition
- set the time derivative to zero and Poisson returns

ORDER IN TIME → CONDITION COUNT

$\frac{\partial u}{\partial t}$: first order $\Rightarrow$ one initial condition

DIFFUSION SMOOTHS



sharp features decay fastest — the opposite of the wave equation

The Space–Time Domain

DOMAIN

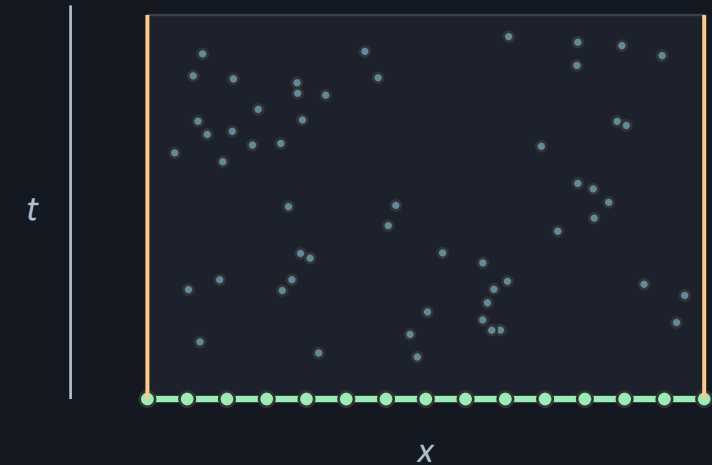
$$(x, y, t) \in \Omega \times (0, t_{end}] \subset \mathbb{R}^3$$

- a PINN does not march through time — time is another coordinate
- the network is trained on the whole space-time block at once
- a genuine departure from FDM and FEM, where each step depends on the last
- **the advantage** — the solution is continuous in t , evaluable at any instant
- the risk is causality: nothing requires early times to be right first

INITIAL CONDITION

$$u(x, y, 0) = f_{IC}(x, y)$$

$\Omega \times \text{TIME}$, SAMPLED AS ONE BLOCK



green: IC at $t = 0$ · amber: DBC for all t

cyan: interior collocation, scattered through space AND time

Hard-Enforcing the Initial Condition

TRIAL SOLUTION ENFORCING BOTH THE IC AND THE DIRICHLET BC

$$u_{trial}(x, y, t) = (1 - t) f_{IC}(x, y) + t x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}(x, y, t)$$

- **read the two terms separately** — at $t = 0$ the second vanishes
- so the initial condition is returned exactly, for any weights
- and for $t > 0$ the spatial factor still vanishes on the boundary
- two loss terms disappear at once
- **the $(1 - t)$ factor assumes t in $[0, 1]$** — non-dimensionalise the time axis

TERM BY TERM

$$(1 - t) f^{IC}(x, y)$$

$$= f^{IC} \text{ at } t = 0$$

$$= 0 \text{ at } t = 1$$

$$t \cdot x(1 - x)y(1 - y)$$

$$= 0 \text{ at } t = 0$$

$$= 0 \text{ on } \partial\Omega, \text{ all } t$$

$$\mathcal{N}_{\hat{w}}(x, y, t)$$

free everywhere else

each factor switches off exactly one condition

Loss Construction and the Causality Caution

SOFT ENFORCEMENT — THREE TERMS

$$\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{DBC} + \mathcal{L}_{IC}$$

HARD ENFORCEMENT — ONE TERM

$$\mathcal{L} = \mathcal{L}_{PDE}$$

- with soft enforcement the three terms compete
- and the initial condition is the one most often lost
- **it occupies a single slice of a space-time domain** — a sampling imbalance, not a pathology
- **two fixes, in order** — hard-enforce it, or weight the early times

The parabolic lesson: when a condition lives on a lower-dimensional slice of the domain, penalising it is structurally weak. Build it into the ansatz instead.

The Wave Equation: Second Order in Time

GOVERNING EQUATION

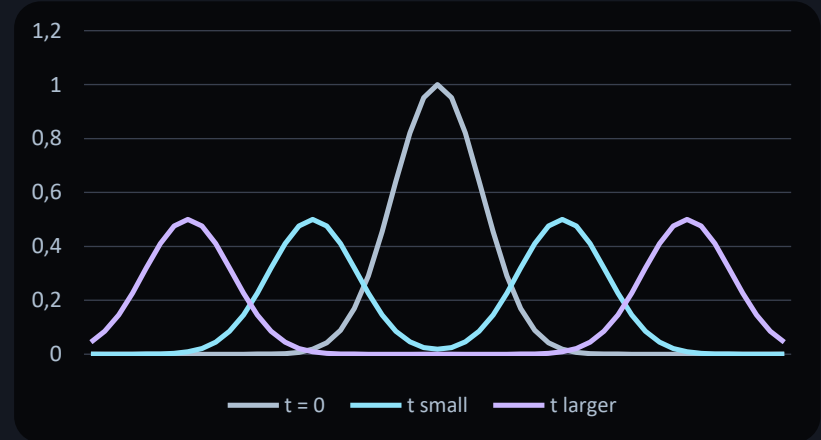
$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + f(x, y, t)$$

- the time derivative is now second order
- and that single change alters the mathematics completely
- sharp features persist instead of decaying
- **solutions oscillate** — which is what spectral bias finds hardest
- and two initial conditions are now required, not one

ORDER IN TIME → CONDITION COUNT

$\frac{\partial^2 u}{\partial t^2}$: second order $\Rightarrow$ two initial conditions

WAVES PRESERVE



amplitude is retained; the pulse splits and travels

Why Two Initial Conditions Are Needed

INITIAL DISPLACEMENT AND INITIAL VELOCITY

$$u(x, y, 0) = u_0(x, y), \quad \dot{u}(x, y, 0) = v_0(x, y)$$

- a plucked string released from rest, against one given a push
- the shape at $t = 0$ is identical; the motion is not
- so displacement alone does not determine the solution
- without the velocity term the problem is genuinely under-determined
- the velocity comes from autograd, like any other derivative

SAME SHAPE, DIFFERENT FUTURES

released from rest

$v_0 = 0$



given a push

$v_0 \neq 0$



grey: $t = 0$, identical in both cases

Periodic Activations for Oscillatory Solutions

ACTIVATION CHOICE

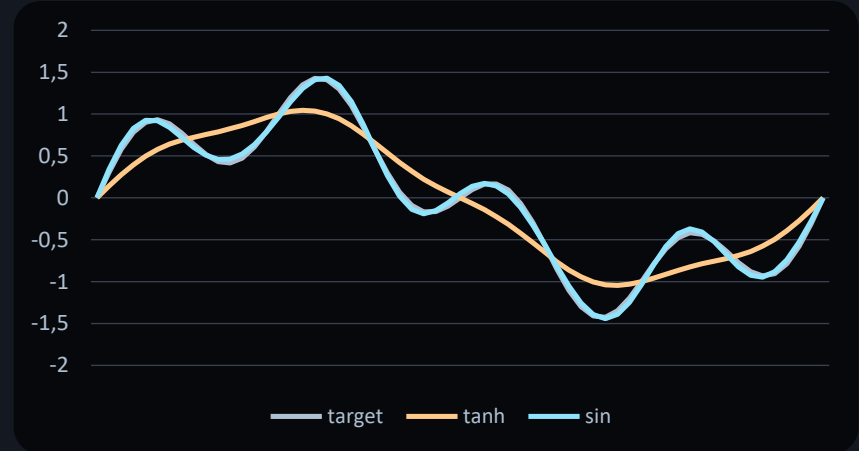
$$\sigma(z) = \sin(z) \quad \text{or} \quad \cos(z) \quad \text{vs.} \quad \tanh(z)$$

- the inductive bias should match the solution
- **a wave solution is a sum of sinusoids** — so build the network from sinusoids
- this is the direct remedy for spectral bias
- SIREN scales the first layer and uses a matched initialisation
- **and the derivative of a sinusoid is a sinusoid** — every order stays conditioned

SIREN FIRST LAYER

$$\phi(\mathbf{x}) = \sin(\omega_0 \mathbf{W}\mathbf{x} + \mathbf{b}), \quad \omega_0 \approx 30$$

FITTING AN OSCILLATORY TARGET



Wave Loss Construction

SOFT ENFORCEMENT — FOUR TERMS

$$\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{DBC} + \mathcal{L}_{IC} + \mathcal{L}_{IV}$$

TRIAL SOLUTION

$$u_{trial}(x, y, t) = (1 - t) f_{IC}(x, y) + t x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}(x, y, t)$$

- **four competing terms** — the worst balancing problem in this lecture
- which is why the wave equation is the standard stress test
- the trial solution hard-enforces the initial displacement and the boundary
- a second-order-in-time residual costs two extra autograd passes

The hyperbolic lesson: an extra order in time costs an extra condition, an extra loss term, and an extra pair of autograd passes. Nothing about it is free.

Helmholtz: When the Eigenvalue Is Also Unknown

GOVERNING EQUATION

$$\nabla^2 u + \lambda u = 0, \quad \lambda = k^2$$

- structurally this is Poisson with the source replaced by lambda times u
- so the unknown appears on both sides
- two things are unknown at once** — the mode shape and the scalar
- only certain pairs satisfy the equation
- it arises whenever you separate variables in the wave or heat equation

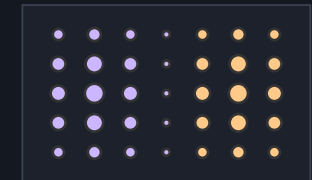
WHERE IT COMES FROM

$$u(x, y, t) = \sum_{m,n} a_{mn}(t) \phi_{mn}(x, y)$$

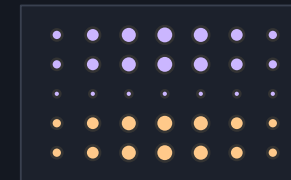
FIRST FOUR MODES OF A SQUARE



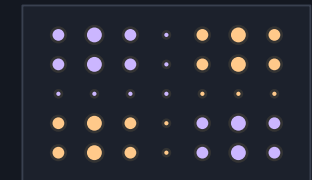
(1,1) $\lambda = 2\pi^2$



(2,1) $\lambda = 5\pi^2$



(1,2) $\lambda = 5\pi^2$



(2,2) $\lambda = 8\pi^2$

The Trivial-Solution Trap

THE SOLUTION THE OPTIMISER WILL FIND FIRST

$$u \equiv 0, \quad \lambda = 0$$

Λ JOINS THE TRAINABLE PARAMETERS

$$\lambda \text{ trainable}, \quad \lambda \geq \lambda_{\min}$$

- u identically zero with λ zero satisfies everything, exactly
- gradient descent will find it, reliably and quickly
- **left unaddressed this is not an occasional failure** — it is the expected outcome
- **two safeguards are needed together** — a normalisation, and a floor on λ

A zero residual does not mean a correct answer. Always ask what else satisfies your loss — the optimiser certainly will.

Mode Orthogonality

ANALYTIC MODES OF THE UNIT SQUARE

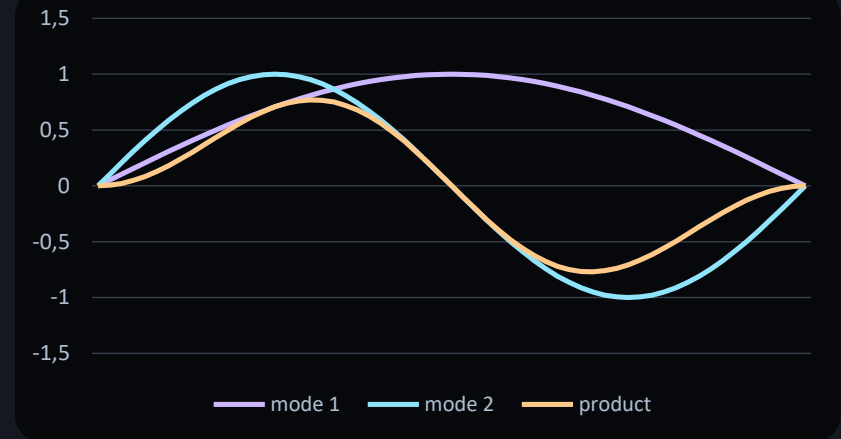
$$u_{m,n}(x, y) = 2\sin(m\pi x)\sin(n\pi y), \quad \lambda_{m,n} = (m^2 + n^2)\pi^2$$

INNER PRODUCT — MODES ARE MUTUALLY ORTHOGONAL

$$\langle u_{m,n}, u_{p,q} \rangle = \int_0^1 \int_0^1 u_{m,n} u_{p,q} dx dy = \delta_{mp} \delta_{nq}$$

- distinct eigenfunctions are orthogonal under the L2 inner product
- which is what makes separation of variables work
- for a PINN it is the tool that prevents mode collapse
- without it every run converges to the lowest mode

ORTHOGONALITY, VISUALLY



the product has equal positive and negative area: the integral is zero

Enforcing Orthogonality: Projection or Penalty

GRAM-SCHMIDT PROJECTION — HARD ENFORCEMENT

$$u \leftarrow u_{raw} - \sum_{k < K} \frac{\langle u_{raw}, u_k \rangle}{\langle u_k, u_k \rangle} u_k$$

PENALTY — SOFT ENFORCEMENT

$$\mathcal{L}_{orth} = \sum_{k < K} \langle u_{\hat{w}}, u_k \rangle^2$$

PROJECTION

Subtract the components along all previously found modes before the residual is evaluated.
Orthogonality is then exact, for any $\hat{w}$.
No weight to tune. This is Liu's method.

PENALTY

Add the squared inner products as a loss term with a weight w_o .
Orthogonality only approximate, and it competes with the PDE residual.
Simpler to code; easier to get wrong.

The same choice as Dirichlet enforcement in L7.1, in a new guise: build the constraint into the ansatz, or pay for it in the loss. Liu projects — and so should you.

Normalisation and the Full Loss

NORMALISATION TERM

$$\mathcal{L}_{norm} = (\|u_{\hat{w}}\|^2 - 1)^2$$

COMPOSITE LOSS FOR AN EIGENVALUE PROBLEM

$$\mathcal{L} = \mathcal{L}_{PDE} + w_o \mathcal{L}_{orth} + w_n \mathcal{L}_{norm}$$

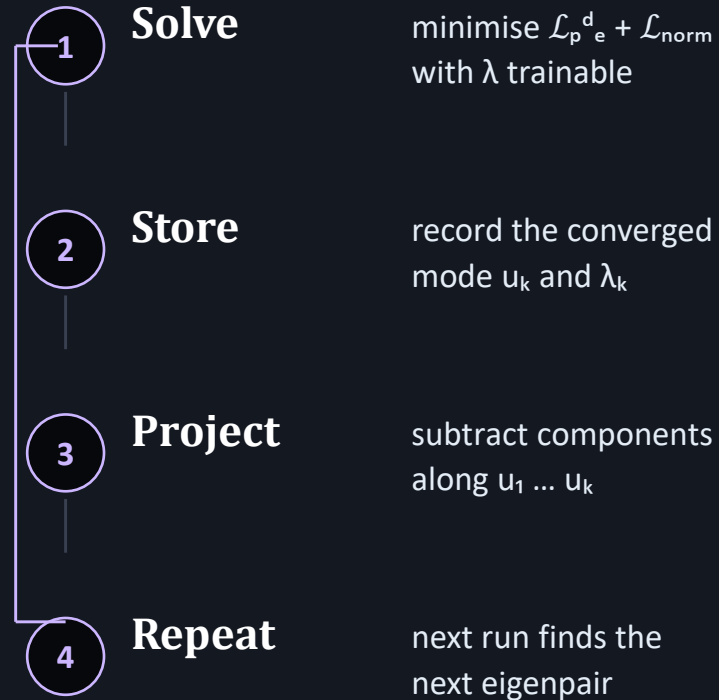
- an eigenfunction is defined only up to a scalar multiple
- so fix the norm, and the eigenvalue becomes comparable across runs
- the norm is an integral, so it needs quadrature weights
- on an irregular domain that is the T3 machinery again

Sanity check for students: on the unit square the lowest eigenvalue must come out at $\lambda = 2\pi^2 \approx 19.74$. If your PINN returns something near zero, the trivial solution has won.

Sequential Training: One Mode at a Time

- ▶ Modes are found in order. Each run solves for one eigenpair, then stores the converged mode and moves on.
- ▶ The stored modes are what the projection step subtracts, so run k is constrained to be orthogonal to everything found in runs 1 through $k-1$.
- ▶ The lowest mode is found first because it is the smoothest — spectral bias, working for you rather than against you for once.
- ▶ Errors compound: an inaccurate mode k pollutes the projection for every mode after it. Converge each one properly before continuing.
- ▶ Practical note: reuse the converged weights of mode k as the initialisation for mode $k+1$. It is a better starting point than random, and the run converges markedly faster.

THE LOOP



Burgers' Equation: Nonlinearity Enters

GOVERNING EQUATION

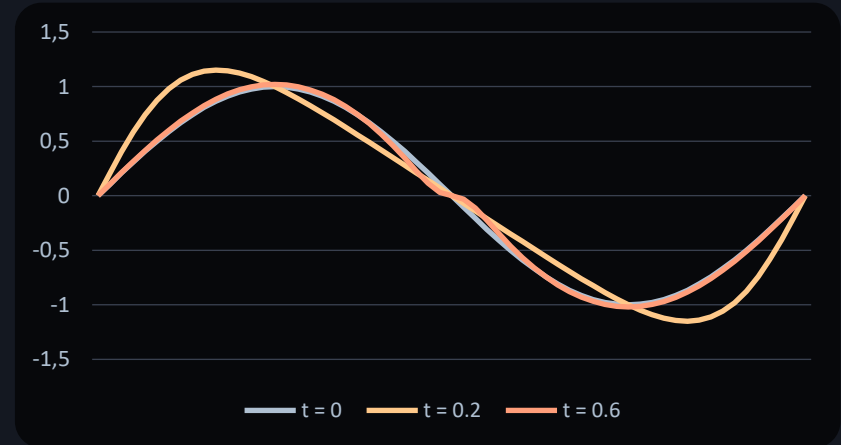
$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad \nu = \frac{0.01}{\pi}$$

- ▶ The first nonlinear PDE in the course: the convection term multiplies the unknown by its own derivative.
- ▶ It is the standard one-dimensional model for the balance in Navier–Stokes between nonlinear convection and viscous diffusion — which is why it is the gateway to the fluid-flow material in L9.
- ▶ At $\nu = 0.01/\pi$ the viscous term is barely enough to keep the solution smooth. A near-discontinuity forms at $x = 0$ by $t \approx 0.4$.
- ▶ Not in Liu's book. This treatment follows the Raissi reference implementation, which uses this problem as its flagship benchmark.

INITIAL AND BOUNDARY CONDITIONS

$$u(0, x) = -\sin(\pi x), \quad u(t, -1) = u(t, 1) = 0$$

SOLUTION STEEPENING IN TIME



a smooth sine steepens into a near-shock at $x = 0$

The Nonlinear Convection Term

LINEAR TRANSPORT VERSUS NONLINEAR CONVECTION

$$c u_{,x} \quad \text{versus} \quad u u_{,x}$$

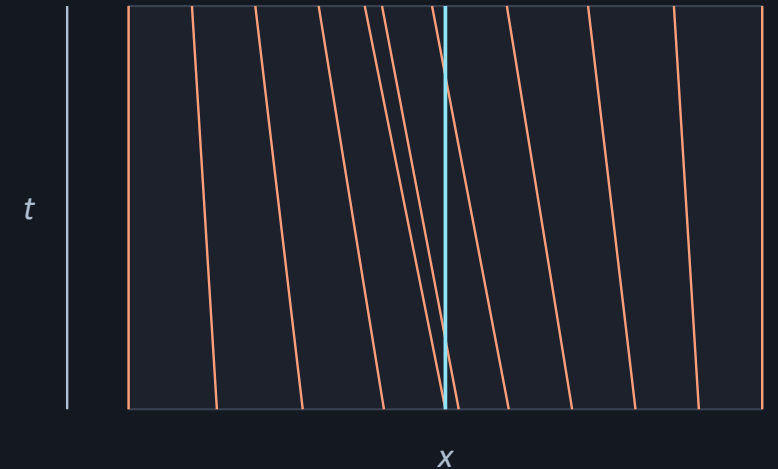
- ▶ In linear transport every point of the profile moves at the same speed c , so the shape is preserved and translated.
- ▶ In Burgers the transport speed is the solution itself. Tall parts of the profile move faster than short parts, so the wave leans forward and steepens.
- ▶ Characteristics therefore converge instead of running parallel. Where they would cross, the inviscid equation has no single-valued solution — a shock. Viscosity ν smooths this into a steep but finite layer.
- ▶ For the PINN, none of this changes the machinery: $u u_x$ is just a product of two quantities the graph already holds. The residual stays differentiable and autograd handles the nonlinearity without comment.

CHARACTERISTIC SPEED

$$\frac{dx}{dt} = u \quad \Rightarrow \quad \text{characteristics intersect}$$

CHARACTERISTICS IN THE X-T PLANE

shock forms here



lines converge — information piles up at a single point

Why Shocks Are Hard for a PINN

SPECTRAL BIAS, SHARPENED

A near-discontinuity is the highest-frequency feature a solution can have. It is precisely what gradient descent learns last, and on a finite budget may never learn at all.

SAMPLING IMBALANCE

The shock occupies a vanishing fraction of the domain. Uniform collocation spends almost every point where the solution is flat and already correct.

LARGE RESIDUAL GRADIENTS

Near the shock u_{xx} is enormous, so the residual and its gradient dominate the loss — destabilising the very region that most needs care.

SMOOTHNESS OF THE ANSATZ

A tanh network is infinitely smooth by construction. Representing a near-discontinuity forces extreme weights, which is an ill-conditioned place for the optimiser to sit.

Remedies, in order: residual-adaptive refinement to place points at the shock; more collocation points overall; and a domain decomposition in time so early, smooth times converge before late, sharp ones.

Loss Construction and the Reference Setup

COMPOSITE LOSS — DATA TERM PLUS RESIDUAL TERM

$$\mathcal{L} = \frac{1}{N_u} \sum_{i=1}^{N_u} |u_{\hat{w}}(t_u^i, x_u^i) - u^i|^2 + \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_f^i, x_f^i)|^2$$

- ▶ Raissi treats the initial and boundary conditions together as a single data set of $N_u = 100$ points, so the loss has two terms rather than four.
- ▶ The residual is enforced at $N_f = 10\,000$ collocation points drawn by Latin hypercube over the space–time rectangle — a hundred times as many as the data points.
- ▶ Architecture: 8 hidden layers of 20 tanh neurons, Xavier initialisation, inputs normalised to $[-1, 1]$. Optimiser: L-BFGS-B alone, to a tolerance at machine epsilon.
- ▶ Both conditions are soft here. A trial solution could hard-enforce $u(t, \pm 1) = 0$, but the initial condition $-\sin(\pi x)$ is harder to build in.

VISCOSITY

$$\nu = \frac{0.01}{\pi} \approx 3.18 \times 10^{-3}$$

REFERENCE CONFIGURATION

data points N_u	100
collocation N_f	10 000
hidden layers	8
neurons per layer	20
activation	tanh
optimiser	L-BFGS-B

The Inverse Variant: Discovering the Equation

TWO COEFFICIENTS PROMOTED TO TRAINABLE PARAMETERS

$$f := u_t + \lambda_1 u u_x - \lambda_2 u_{xx}$$

- ▶ λ_1 and λ_2 replace the known coefficients. They are initialised at 0 and e^{-6} , far from the truth, and recovered by the same gradient step that trains the weights.
- ▶ The true values are $\lambda_1 = 1$ and $\lambda_2 = \nu = 0.01/\pi \approx 3.18 \times 10^{-3}$. Both are recovered to within about one percent from noisy scattered data.
- ▶ What has been discovered is not just two numbers but the balance between them — how much of the dynamics is convection and how much is diffusion. That is a statement about the physics.
- ▶ This is the template for every parameter-identification problem in the course: an unknown diffusivity in L8, an unknown viscosity in L9, an unknown line impedance in L11.

The forward and inverse problems differ by two lines of code. That asymmetry — trivial for a PINN, expensive for a classical solver — is one of the strongest arguments for the method.

The Five PDEs Compared

	POISSON	HEAT	WAVE	HELMHOLTZ	BURGERS
Type	elliptic	parabolic	hyperbolic	eigenvalue	nonlinear
Inputs	x, y	x, y, t	x, y, t	x, y	x, t
Order in t	—	1st	2nd	—	1st
Conditions	DBC / NBC	DBC + IC	DBC + IC + IV	DBC + norm.	IC + DBC
Hard enforcement	easy	easy	partial	projection	awkward
Activation	tanh	tanh	sin / tanh	sin / tanh	tanh
Main difficulty	geometry	causality	balancing 4 terms	trivial solution	shock

Read across any row to see what a change of PDE type actually costs you. Read down the last row for the reason each one earns a place in this lecture.

Which Loss Terms Appear, and When

	$\mathcal{L}_{p_e}^d$	$\mathcal{L}^{DBc}$	$\mathcal{L}^{NBc}$	$\mathcal{L}^{lc}$	$\mathcal{L}^{lv}$	$\mathcal{L}_{norm}$
Poisson	●	○	○			
Heat	●	○		○		
Wave	●	○		○	●	
Helmholtz	●	○				●
Burgers	●	●		●		

● required ○ present under soft enforcement, removable by a trial solution — not applicable

Every open circle is a design decision, not a fact about the physics.

Pitfalls by PDE Type

Elliptic

Pure Neumann leaves the solution non-unique up to a constant.

Symptom: converged residual, wrong absolute level. Fix: pin one value, or add a mean constraint.

Parabolic

The IC lives on a measure-zero slice and is outvoted by interior points.

Symptom: late times fit, early times do not. Fix: hard-enforce the IC via the trial solution.

Hyperbolic

Four loss terms compete, and oscillatory solutions meet spectral bias head on.

Symptom: smooth envelope, missing detail. Fix: periodic activations, denser sampling.

Eigenvalue

The trivial solution is a global minimum of the residual.

Symptom: $u \rightarrow 0$ and $\lambda \rightarrow 0$. Fix: normalisation term plus a floor on λ .

Nonlinear

Sharp features attract enormous residual gradients while occupying almost no domain.

Symptom: oscillation near the shock. Fix: adaptive refinement, time decomposition.

Each symptom is visible in plots you already produce: per-term loss curves and a residual map over the domain.

From These Five PDEs to the Application Lectures

L8**Heat dissipation**

Poisson + transient heat

Steady and transient conduction in devices and heat sinks — exactly the elliptic and parabolic problems above.

L9**Fluid flow & batteries**

Burgers → Navier–Stokes; coupled transport

Burgers is the convection–diffusion balance in one dimension. DFN battery models couple diffusion with Butler–Volmer kinetics.

L10**Autonomous robots**

Wave + eigenvalue thinking

Vibration and structural modes; physics-informed learning where the constraint is a dynamic model rather than a PDE.

L11**Power grids**

Poisson on a graph Laplacian

DC power flow is a discrete Poisson problem; optimal power flow adds constraints of the kind seen in the Helmholtz loss.

None of these lectures introduces a new PINN method. They change $\mathcal{F}$ and the conditions — which is the entire claim of L7.

Non-Dimensionalisation: Do It First

- ▶ Physical problems arrive with awkward scales: a diffusivity of $10^{-5} \text{ m}^2/\text{s}$, a domain of 10^{-3} m , temperatures of 300 K. Fed to a network with tanh activations and Xavier initialisation, none of these are in a comfortable range.
- ▶ Every reference implementation in this course works on a unit domain with an $O(1)$ solution. That is not a simplification for teaching — it is a necessary conditioning step.
- ▶ Rescale x , t and u so each lies roughly in $[-1, 1]$ or $[0, 1]$. The PDE keeps its form; the coefficients absorb the scaling and usually collapse into one dimensionless group.
- ▶ The trial solutions in this lecture assume it. The factor $(1 - t)$ only enforces an initial condition if t has been scaled so that $t \in [0, 1]$.
- ▶ Raissi normalises inputs inside the network itself — the first operation maps the input box to $[-1, 1]$ before the first affine layer.

BEFORE AND AFTER

QUANTITY	PHYSICAL	SCALED
length	$1.0 \times 10^{-3} \text{ m}$	1
time	$5.0 \times 10^2 \text{ s}$	1
diffusivity	1.4×10^{-5}	0.7
temperature	$300 \rightarrow 340 \text{ K}$	0 $\rightarrow$ 1

A PINN that will not train often has nothing wrong with it except its units.

How to Know Your Solution Is Right

MANUFACTURED SOLUTION

Choose u first, substitute it into $\mathcal{F}$ to derive the source f , then solve and compare. Every PDE in this lecture has one — it is how the reference errors were measured.

KNOWN EIGENVALUES

For Helmholtz on the unit square, $\lambda_{mn} = (m^2 + n^2)\pi^2$ exactly. The lowest is $2\pi^2 \approx 19.74$. Any large deviation is a bug, not a discretisation error.

CONSERVATION CHECKS

For the heat equation with insulated boundaries, total energy must be conserved. Integrate the solution and watch the drift — it bounds your error from below.

CONVERGENCE WITH N

Refine the collocation set and confirm the error falls. If it plateaus, the limit is the network or the optimiser, not the sampling.

A PINN gives no error estimate of its own. Unlike FEM there is no mesh to refine and no residual bound — verification is entirely your responsibility.

Ex_08 — Heat and Wave on the Unit Square

TASKS

- 1 Extend `pinn_core` to three inputs (x, y, t) and solve the transient heat equation with a soft IC.
- 2 Re-solve it with the IC hard-enforced by the $(1-t)$ trial solution. Compare the error at $t = 0$.
- 3 Solve the wave equation with both IC and IV. Then remove $\mathcal{L}^{IV}$ and explain what you observe.
- 4 Repeat the wave run with sin activations and compare against $\tanh$ at equal parameter count.
- 5 Report relative L_2 against the analytic solution for each case.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why does the heat equation need one IC and the wave equation two? sl. 13, 14

Why is hard-enforcing the IC more effective than weighting it? sl. 11, 12

What goes wrong if the eigenvalue problem has no normalisation term? sl. 18, 21

Why does Burgers need adaptive sampling when Poisson does not? sl. 24, 25

Which PDE type is your own project problem, and what follows from that? sl. 3, 28

Built on the same `pinn_core.py` as Ex_07 — only the residual, the conditions and the trial solution change.

Questions

TEN TO TEST WHETHER TODAY LANDED

- what does the discriminant tell you, and why does it matter for the loss?
- why does Poisson need no initial condition?
- what does a PINN do differently from a time-marching solver?
- why does the wave equation need two initial conditions?
- what does a sinusoidal activation buy on an oscillatory problem?
- why is u identically zero a problem for Helmholtz and not for Poisson?

AND FOUR MORE

seven

how does hard enforcement handle an initial condition?

eight

why does a pure Neumann problem need an extra constraint?

nine

what does mode orthogonality prevent?

ten

your solution converged. How do you know it is right?

THE ONE WITH NO SETTLED ANSWER

Causality is not enforced by the loss. Should it be, and at what cost?

BEFORE L8.1

Ex_07.2 notebooks 01 to 03. Bring the wave-equation loss curves.

Key Takeaways

- 1 Type dictates conditions**

The discriminant tells you whether a problem is elliptic, parabolic or hyperbolic, and that determines which conditions make it well posed — and therefore which loss terms you must carry.
- 2 Order in time costs terms**

First order in time needs one initial condition; second order needs two. The wave equation carries $\mathcal{L}^{lv}$ for a reason, and omitting it does not simplify the problem — it breaks it.
- 3 Hard enforcement recurs**

A trial solution for Dirichlet, a time factor for the initial condition, Gram–Schmidt projection for orthogonality. The same idea in three guises: build the constraint in, do not pay for it in the loss.
- 4 A zero residual is not enough**

Helmholtz makes it explicit: $u \equiv 0$ minimises the residual perfectly. Always ask what else your loss admits.
- 5 Nonlinearity is free; shocks are not**

Autograd handles uu_x without comment. What costs you is the sharp feature it creates, and that is a sampling and conditioning problem.

Next — L8: the heat equation in earnest, on real device geometry.